Worksheet & Answer Key

The Earth's atmosphere bends light as it enters our eyes, a phenomenon known as atmospheric refraction. This is because air molecules have a slightly different density than the surrounding air, causing light to bend and change direction. As light travels through the air, it passes through layers of different densities, with each layer having its own unique refractive index. This bending effect allows us to see far-off objects more clearly, as the light from those objects has been refracted or bent towards our eyes. In addition to this correctional role, atmospheric refraction also plays a part in the spectacular displays of colorful sunsets and sunrises we enjoy on clear days.

Word Bank:

curves, layers, density, bends, refracts, clears, objects, surfaces

Fill in the blanks:

1. A_____ bends light as it enters our eyes.
2. The Earth's atmosphere _____ light as it enters our eyes.
3. As light travels through the _____, it passes through layers of different densities.
4. This bending effect allows us to see far-off _____ more clearly.
5. The refractive index changes as we move through _____.
6. In addition to this correctional role, atmospheric refraction also plays a part in the spectacular displays of colorful _____.
7. Light from far-off _____ has been refracted or bent towards our eyes.
8. This is because air molecules have a slightly different _____ than the surrounding air.

■ Answer Key (for teachers only):

1. bends
2. curves
3. layers
4. bends
5. refracts
6. clears
7. objects
8. density